

Waterman

Given: Object set O and an additive metric on O .

$$O = \{A, B, C, D, E\}$$

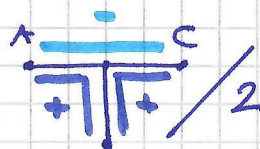
	A	B	C	D	E
A		3	7	10	7
B			8	11	8
C				9	6
D					5
E					

- ① Pick two arbitrary objects and connect them by an edge whose weight corresponds to the entry in the matrix

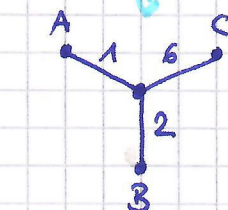


- ①.1 Pick a pair of leaves and ②.1 compute the weight of the new edge which connects a new object

We want to connect B.



$$\text{③.1} \quad \frac{d_{AB} + d_{CB} - d_{AC}}{2} = \frac{3 + 8 - 7}{2} = 2$$



Split edge AC, insert a node and attach the new edge to that node

- ①.2 We could also pick AC (or A, B) for D. But we choose BC to insert D.

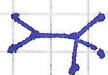
- ②.2 Then we compute

$$\frac{d_{BD} + d_{CD} - d_{BC}}{2} = \frac{11 + 9 - 8}{2} = 6$$

- ③.2 And insert D into the tree



- ①.3 We cannot choose B, C to insert E, as that would lead to:



Hence, we choose a different edge: C, D

- ②.3 We compute the new length

$$\frac{d_{CE} + d_{DE} - d_{CD}}{2} = \frac{8 + 5 - 11}{2} = 1$$

- ③.3 And insert the node into the tree:

